TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 4 THEME · WORKBOOK .RULES

Lab 4 — Create and Test Workbook .Rules for Water Service Reporting

Maps to	LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence.
Duration	30 minutes
Tools	Microsoft 365 Copilot in Excel · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-4

Goal

Turn Northstar's workbook conventions into a visible .Rules worksheet. Keep one rule per cell in column A, include a formula-driven reporting-date rule, test the output, and verify that the sheet travels with a copied workbook. A .Rules sheet is guidance for Copilot, not Excel formula enforcement or a security boundary.

What You Will Do

Turn Northstar's workbook conventions into a visible .Rules worksheet. Keep one rule per cell in column A, include a formula-driven reporting-date rule, test the output, and verify that the sheet travels with a copied workbook. A .Rules sheet is guidance for Copilot, not Excel formula enforcement or a security boundary.

What You Will Build

Northstar_Rules_Workbook.xlsx with visible .Rules column A, a checked Rules_Test output and a copied-workbook sharing check.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 3 scope comparison; verified Lab1 clean source or supplied checkpoint.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Open the clean workbook, save as Northstar_Rules_Workbook.xlsx and inspect Control. In Copilot use Add work content > Create workbook rules if offered; otherwise insert a sheet manually and rename it exactly .Rules, including the leading dot. Keep it visible. Rules are fully supported only in English.

PROMPT / CHECK — copy and adapt

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EXCEL - CHAT ONLY:
Inspect this Northstar workbook. Identify the source table, fixed reporting date, status vocabulary and current owners. List workbook-specific conventions that belong in .Rules rather than account Personalisation. Do not edit.
```

Step 2. Create the .Rules lines with the one-shot Allow editing prompt below (or copy them from the prompts page): one heading or rule per column-A cell; no data in other columns is required for recognition. Check source preservation, backlog/overdue/rate definitions, S$ display, colours, scope and controlled replacements.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
Add a visible worksheet named exactly .Rules (with the leading dot) as the last tab; if it exists, clear column A and reuse it. Fill column A only, one cell per line below, entering only the text after each "A#: " label. Do not change any other sheet.
A1: ## SOURCE AND SCOPE
A2: Preserve Work_Orders, tblWorkOrders, Technicians, current Assigned_To values and source IDs.
A3: Use the reporting date from Control!B2. State the selected Site, Job_Category, Status and Assigned_To scope beside every report.
A4: Backlog excludes Closed and Cancelled. Overdue backlog has Due_Date strictly before Control!B2.
A5: On-time closure rate uses Closed rows only, with Closed_Date <= Due_Date. Display N/A when no Closed rows remain.
A6: ## FORMATS AND VISUALS
A7: Display recorded Cost_SGD as S$ with two decimals; keep underlying values exact. Show dates as DD-MMM-YYYY.
A8: Use white backgrounds, navy titles, blue and teal accents, readable labels and named units; never use colour alone.
A9: ## DASHBOARD LAYOUT
A10: Dashboard sheet: title in A1; Site, Job_Category, Status and Assigned_To dropdowns in B2:B5 (All plus source values); the six measures in B7:B12; data-as-at date and selected scope in D2.
A11: Six KPI tiles in D4:I7, one column per tile: title in row 4, large value in row 5, short definition in row 6 and "Checks: PASS" in row 7.
A12: Charts in a grid below the tiles: work orders by status and by job category side by side, then a full-width stacked bar by site (overdue backlog, other backlog, Closed, Cancelled).
A13: Slicers for Site, Job_Category, Status and Assigned_To on the right of the tiles, connected to tblWorkOrders. KPI formulas use SUBTOTAL visibility so slicers and dropdowns both filter every tile and chart.
A14: Put chart summaries, chart-title formulas and KPI checks on Dashboard_Summaries. Never merge cells. Hide gridlines on Dashboard.
A15: ## DASHBOARD STYLE
A16: Tiles: white bold text on solid navy #1F4E79 (totals and rates), blue #2E75B6 (backlog, overdue) or teal #2A9D8F (Closed, cost).
A17: Every chart title states the finding from a cell formula, e.g. "Changi Workshop has the most overdue backlog (18 of 50)".
A18: Show data labels on bars; no gridlines; rates as % with one decimal. Each chart total must equal B7.
A19: ## OUTPUT AND REVIEW
A20: Use structured table references. Preserve the visible .Rules sheet and unrelated sheets, charts and Prompt_Log.
A21: Keep current owners separate from unapproved assignment proposals. Missing facts and approvals remain UNKNOWN.
A22: Ask before replacing named output sheets. Reconcile every count, rate and chart against source rows; do not duplicate outputs on rerun.

EXCEL - PLAN:
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Review the visible .Rules column A for this Northstar workbook. Identify ambiguous
instructions or conflicts with the actual source and Control. Propose concise corrections,
keeping one idea per cell. Do not edit the source or invent precedence between rules,
preferences and Skills.
```

Step 3. After reviewing any correction, update only .Rules. Add a formula in the next empty column-A cell: ="Reporting date: "&TEXT(Control!B2,"yyyy-mm-dd"). Recalculate and confirm it displays Reporting date:2026-09-30. Copilot reads the formula result on the next prompt; this does not rewrite dashboard values by itself.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
Apply only my reviewed .Rules corrections. Keep the sheet visible, one instruction per
column-A cell, and retain the reporting-date formula. Preserve source data, Control and
unrelated sheets.
```

Step 4. Request a Rules_Test summary without restating the formatting rules. Check counts against Control, source references, displayed date/scope, S$ units and proposal wording. The known baseline is120 total,60 backlog,50 overdue and 30/48 on-time Closed.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
Create Rules_Test for Northstar Water Services from tblWorkOrders using the visible workbook
.Rules. Include total, backlog, overdue, on-time Closed rate and recorded cost with
formula/source evidence. Show date and All scope. Keep source owners and values unchanged.
```

Step 5. Save As Northstar_Rules_Shared_Copy.xlsx. Open the copy and confirm .Rules is still present and visible in column A; you have not exported your account preferences. If testing a hidden sheet, use only a sandbox copy, hide .Rules, record that hidden rules are not used, then unhide before continuing. Do not claim native rule use when the feature is unavailable.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
In this copied Northstar workbook, identify the visible .Rules instructions and reporting-date
formula result. Explain which guidance travels with the workbook and which account preferences
do not. Verify Rules_Test against Control without changing records.
```

Step 6. Record the sheet name, visibility, column, formula result, actual output checks and copied-workbook check in the Evidence_Log sheet. Carry the rules-enabled workbook forward. If a prompt, Skill and rule disagree, ask for clarification and verify the output; no universal precedence is assumed.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Compare Personalisation, .Rules and a reusable Skill for Northstar reporting. Place currency
preferences, this workbook's backlog/date rules and the repeatable Pareto workflow in the
right scope. Identify conflicts requiring clarification and the checks a person still must
perform.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-4

#	Purpose	Prompt
1	Scope	State actual source, report date and selected scope before acting.
2	Verify	Explain the source/formula checks and preserve unknowns and original records.

Test It

Check	Expected result
Recognition	Exactly .Rules; visible worksheet; instructions in column A, one idea per cell.
Formula rule	Reporting-date formula recalculates and displays2026-09-30; no source values changed.
Output controls	120 total;60 backlog;50 overdue;30/48 on-time;S$35,827.50 at All scope.
Workbook copy	Visible .Rules travels with the copy; account preferences are not exported.

Checkpoint

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?